

Rajat Savdekar

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EDUCATION

University of Southern California, Los Angeles, CA Aug 2024 – May 2026
Master of Science in Computer Science
GPA: 3.63
Coursework: Algorithms, Database Systems, Machine Learning, Web Technology, Distributed Systems

University of Mumbai, Mumbai, India Aug 2019 – May 2023
Bachelor of Engineering in Computer Engineering
CGPA: 9.4/10
Coursework: Operating Systems, Computer Networks, Distributed Systems, Natural Language Processing

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, SQL, React, Next.js, HTML, Tailwind CSS
AI & Agents: OpenAI API, LangChain, RAG, vector search, pgvector, FAISS, ChromaDB, prompt engineering
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, WebSockets, gRPC
Cloud & Infrastructure: AWS (EC2, S3, CloudWatch), Docker, Terraform, GitHub Actions, CI/CD

EXPERIENCE

Software Developer Aug 2026 – Present
Easley-Dunn Productions, Inc. – Mediverse Torrance, CA

- Reduced AI feedback latency from 22 to 4 minutes by building a **production LLM integration** with a 4-bit quantized 7B model, implementing **prompt engineering** and **hallucination reduction** to ensure reliable responses under **concurrent load** within a strict 4 GB RAM constraint.
- Sustained locked **72 FPS** on severely memory-constrained hardware by profiling and resolving race conditions across 6 real-time modules, enforcing thread-safe execution with **CPU/GPU optimization** and performance benchmarking before every deploy.

Software Engineer May 2023 – May 2024
Jalgaon Fruit Sales Cooperative Ltd (JFSS) Jalgaon, India

- Doubled peak harvest write throughput by decomposing a **legacy PHP monolith** into **Python Flask microservices** on **PostgreSQL**, eliminating MySQL table-lock freezes and cutting query latency 40% across 1,000+ daily truck transactions.
- Containerized services with **Docker**, deployed to AWS EC2 via **GitHub Actions CI/CD**, and instrumented with **CloudWatch** monitoring and **pytest** integration tests enforcing 80% coverage on every PR.
- Designed a **production RAG pipeline** using **LangChain**, **OpenAI embeddings**, and **ChromaDB** over 150K+ procurement records, enabling **natural-language semantic search** and cutting average lookup time from 3 minutes to under 20 seconds.
- Engineered a **production LLM inference pipeline** using **GPT-4 Turbo** and async processing with **exponential backoff** on rate limit errors and **Redis caching**, eliminating a 2-hour daily manual reconciliation process across 1,000+ truck transactions.

PROJECTS

MockPad – Real-Time Collaborative Code Editor [Live] [GitHub]
TypeScript, Node.js, React, WebSockets, Yjs CRDTs, OpenAI Whisper, MongoDB, LevelDB May 2026

- Scaled to **10–20 daily active users** on a \$6/month droplet by combining a Yjs CRDT sync layer over **WebSockets** with an LRU eviction policy offloading idle rooms to **LevelDB**, sustaining **100% uptime** under concurrent load.
- Shipped **session replay** with deterministic playback and a **React + TypeScript** frontend on **Monaco Editor** delivering a free, self-hostable alternative to CoderPad for live interviews.

DevStack – AI-Native Developer Infrastructure Platform [GitHub]
TypeScript, Python, Node.js, PostgreSQL, Redis, TimescaleDB, Docker, LangChain, FastAPI Aug 2026

- Architected 8 **microservices** behind an **API gateway** with **JWT**, **RBAC**, sub-5ms feature flag evaluation via **Redis pub/sub** and **WebSocket** push, and a workflow engine using event sourcing for exactly-once execution.
- Built an **LLM eval engine** distributing **isolated Docker sandboxes** via **PostgreSQL SKIP LOCKED** with lease-based crash recovery and **LLM-as-judge** scoring over **RAG-indexed** error logs.
- Shipped an **autonomous CI repair agent** using LangChain that triggers on **GitHub webhook** failures, retrieves context via **semantic search**, and opens fix PRs automatically with structured output.

RESEARCH & HACKATHONS

Publications. *Texture Feature Analysis* – MATLAB tool cited by 10+ papers in IEEE and Springer volumes. [GitHub]

Hackathons. Competed in 6+ events including **OpenAI Build Week**, **CALL-E 2026 Reddit Devvit** [Demo].